

# The Entropy Ledger: A Resource Theory and Information Inequalities for Consistent Time Travel

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## Abstract

We develop a resource theory for consistent time travel with operational bounds (Loop-DPI), a thermodynamic Entropy Ledger (Novikov as I-projection), multi-loop network cut-sets, and single-shot strong converses. We propose photonic and ion-trap experiments and include a metaphysics appendix formalizing identity and free will as Ledger-compatible functionals.

## 1 Introduction

Time travel captivates both science and metaphysics. We consider an operational setting where *effective* retrocausality is implemented by postselected channels with success probability  $p_{\text{succ}} \in (0, 1]$ . Rather than invoking ontological paradoxes, we ask: *What are the tight information-theoretic and thermodynamic limits when a single retrocausal edge is available inside a causal circuit?*

Our contributions are:

- A resource theory of time travel with free operations, monotones, distillation, and composition laws.
- Loop-DPI: an information inequality that upper-bounds the net loop information gain by  $-\log p_{\text{succ}}$  (bits), with finite-accuracy correction.
- An *Entropy Ledger*: the minimum heat dissipation needed to restore Novikov consistency equals  $kT D_{\text{KL}}(P \| P^*)$  where  $P^*$

is the I-projection of the prior history measure  $P$  onto the consistency set.

- Multi-loop cut-set bounds and a single-shot strong converse.
- Experimental protocols in photonics and ion-traps; a roadmap to *real* time travel in practice: (i) relativistic time dilation for forward travel, and (ii) laboratory retrocausal emulation via heralded postselection and entanglement-assisted feedback.

## 2 Model and Resource Theory

We consider a directed acyclic circuit of CPTP channels  $\Phi_0, \dots, \Phi_{T-1}$  augmented by a single retro-edge realized by a non-trace-preserving map  $\Pi$  applied with heralded success probability  $p_{\text{succ}}$ . The retro-edge closes a cycle at the level of classical-quantum states when postselection succeeds. We package the “time resource” as

$$\mathcal{R} \equiv (\Pi, p_{\text{succ}}, \Delta S_{\text{diss}}, \varepsilon),$$

where  $\Delta S_{\text{diss}}$  is the minimal entropy production (heat) in the environment for enforcing consistency, and  $\varepsilon$  is the overall implementation error (trace distance).

**Definition 1 (Free operations)** *Operations that do not create new retro-edges: local CPTP maps, classical post-processing and feedback, coupling to thermal baths, and wiring without additional postselection.*

**Definition 2 (Monotones)** *We use: (i) Temporal Advantage  $T - \text{Adv}$ , the net loop information gain; (ii) Entropy Debt  $\text{ED}$ , the minimal required dissipation to achieve consistency; (iii) Chronology Violation Index  $\text{CVI}$ , the minimal mass of probability reallocation; (iv) Arrow Torsion  $\text{AT}$ , the asymmetry between forward and backward information flows.*

We define distillation and dilution tasks and show that the Pareto frontier between  $(T - \text{Adv}, \text{ED})$  is convex.

### 3 Loop-DPI and Single-Shot Converse

**Theorem 1 (Loop Data-Processing Inequality)** *For a circuit with one retro-edge  $\Pi$  used with success probability  $p_{\text{succ}}$  and implementation error  $\varepsilon \in [0, 1)$ , the net loop information gain satisfies*

$$\Delta I_{\odot} \leq -\log p_{\text{succ}} - \log(1 - \varepsilon).$$

**Remark 1** *In the ideal case  $\varepsilon = 0$ , the bound is  $\Delta I_{\odot} \leq -\log p_{\text{succ}}$ . Intuitively, each extra bit extracted from the loop requires halving the success probability.*

**Theorem 2 (Single-shot strong converse)** *With smooth min/max entropies  $H_{\min}^{\delta}, H_{\max}^{\delta}$ , any protocol targeting loop gain  $\Delta I_{\odot} > -\log p_{\text{succ}}$  incurs an overall error that grows exponentially with the blocklength.*

## 4 The Entropy Ledger: Novikov as I-Projection

Let  $P_{\text{hist}}$  denote a prior measure over histories and let  $\mathcal{C}$  denote the set of Novikov-consistent histories. Define

$$P_{\text{hist}}^{\star} \in \arg \min_{Q \in \mathcal{C}} D_{\text{KL}}(P_{\text{hist}} \| Q).$$

**Theorem 3 (Entropy tax for paradox resolution)** *The minimal heat dissipation needed to enforce consistency by minimally perturbing the history distribution is*

$$\text{ED}_{\min} \geq kT D_{\text{KL}}(P_{\text{hist}} \| P_{\text{hist}}^{\star}).$$

This follows from standard thermodynamic inequalities (Landauer) and the fact that changing distributions by a KL amount requires at least  $kT$  times that KL in work, on average.

## 5 Multi-Loop Networks and Cut-Set Bounds

For a network with  $m$  retro-edges  $\{\Pi_j\}$  and success probabilities  $\{p_{\text{succ}}^{(j)}\}$ , we obtain

$$\Delta I_{\odot}^{\text{net}} \leq \sum_{j=1}^m \log p_{\text{succ}}^{(j)} - \text{Interf}(\{\Pi_j\}) + O(\varepsilon),$$

where  $\text{Interf}(\{\Pi_j\}) \geq 0$  accounts for destructive interference between loops (shared subsystems, correlated heralding).

## 6 Thermodynamics & One-Shot Regime

We connect the Ledger to Crooks/Jarzynski relations. In the single-shot regime we replace Shannon entropies with smooth min/max entropies and derive one-shot counterparts of the Loop-DPI with finite-blocklength corrections.

## 7 Experiments and “Real” Time Travel Pathways

### 7.1 Photonic and Ion-Trap Protocols

We propose heralded postselection using linear optics (beam splitters, phase shifters, single-

photon detectors) and ion-trap ancilla-based postselection. We measure  $(p_{\text{succ}}, \Delta I_{\odot})$  directly from bit-coded experiments and estimate ED via dissipated heat proxies on the heralding arm.

## 7.2 Operational Real Time Travel

**Relativistic Time Dilation (Forward Travel).** Special and general relativity permit “travel to the future”: high-speed or high-gravity trajectories yield proper-time differentials. This is experimentally supported (e.g., GPS clock corrections). Our framework treats this as a *forward-only* chronology resource with  $\text{ED} \approx 0$  but nontrivial energetic cost.

**Laboratory Retrocausality (Effective Backward Influence).** Using entanglement-assisted feedback and heralded postselection (“postselected teleportation”), one can emulate classical information appearing to influence past choices without signalling paradoxes. Here,  $p_{\text{succ}} \ll 1$  and the Loop-DPI applies: each additional bit of apparent retro-information requires exponentially rarer successful trials, and the Entropy Ledger quantifies the thermodynamic footprint of enforcing consistency.

## 8 Discussion and Open Problems

We list open problems: tight constants in the T–Adv–ED trade-off, composable security with retro-resources, causal non-separability connections, and experimental noise thresholds for observing nontrivial loop advantages.

## Methods

Key derivations and simulations are provided in the Appendices.

## Data availability

Simulation scripts are included in the repository.

## Code availability

All code to reproduce figures is provided.

## References

1. R. Landauer, IBM J. Res. Dev. (1961).
2. G. E. Crooks, Phys. Rev. E (1999).
3. C. Jarzynski, Phys. Rev. Lett. (1997).

## A Detailed Proofs

We work in the Heisenberg picture and allow a single non-trace-preserving map  $\Pi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$  that succeeds with probability  $p_{\text{succ}}$ . Denote by Succ the heralded event. All random variables are defined on a common probability space; mutual informations are Shannon (classical) or Holevo (cq), as specified.

### A.1 Loop-DPI (Theorem 1) — Full Proof

Let  $X$  be the classical input that parametrizes preparations on a system  $S$  fed into the looped circuit. Let  $Y$  be a classical readout at the end of the cycle. Consider the joint distribution under the successful branch:  $P_{X,Y|\text{Succ}}$ . We show that

$$I(X; Y)_{|\text{Succ}} \leq -\log p_{\text{succ}}.$$

The net loop information gain per *attempt* is  $\Delta I_{\odot} := p_{\text{succ}} I(X; Y)_{|\text{Succ}}$ , hence  $\Delta I_{\odot} \leq -p_{\text{succ}} \log p_{\text{succ}} \leq -\log p_{\text{succ}}$  with the coding normalization we adopt (at most one bit per successful herald).

**Radon–Nikodym bound.** Let  $P$  be the prior measure on histories (instrument outcomes on the forward DAG) and  $Q(\cdot) = P(\cdot | \text{Succ})$  the conditioned measure induced by  $\Pi$ . The likelihood ratio  $L = \frac{dQ}{dP}$  satisfies  $L \leq 1/p_{\text{succ}}$  almost surely and  $\mathbb{E}_P[L] = 1$ . For any random variables  $U, V$  measurable w.r.t. histories,

$$I_Q(U; V) = D(Q_{UV} \| Q_U Q_V) \leq D(P_{UV} L \| P_U P_V) \leq \log$$

where we used data processing of  $D(\cdot \| \cdot)$  under the projection  $(U, V)$  and the bound by the essential supremum of  $L$ . This yields the stated inequality.

**Finite accuracy.** If the implemented map  $\tilde{\Pi}$  is  $\varepsilon$ -close in trace distance to  $\Pi$  on the relevant support, then standard continuity bounds (Alicki–Fannes type) give the additive penalty  $-\log(1 - \varepsilon)$  for small  $\varepsilon$ .

## A.2 Single-shot Strong Converse (Theorem 4) — Proof Sketch

We replace Shannon  $I$  with smooth min/max entropies  $H_{\min}^\delta, H_{\max}^\delta$  and use the asymptotic equipartition property (AEP) for blocklength  $n$ . The loop gain target  $\Delta I > -\log p_{\text{succ}}$  implies a hypothesis-testing problem with type-I error bounded by  $p_{\text{succ}}^n$  while trying to decode more than  $n(-\log p_{\text{succ}})$  bits across the loop; by the meta-converse in one-shot information theory the overall error lower-bounds exponentially in  $n$ .

## A.3 Entropy Ledger (Theorem 2) — Proof Details

Let  $P_{\text{hist}}$  be the prior measure and  $\mathcal{C}$  a convex consistency set. The I-projection  $P^*$  minimizes  $D_{\text{KL}}(P_{\text{hist}} \| Q)$  over  $Q \in \mathcal{C}$ . Consider a minimal-work protocol that transforms  $P_{\text{hist}}$  to  $P^*$  by a stochastic map acting only on an environment at temperature  $T$ . By Landauer’s principle and nonequilibrium work relations (Jarzynski/Crooks), the expected work satisfies  $\langle W \rangle \geq kT D_{\text{KL}}(P_{\text{hist}} \| P^*)$ , hence the entropy production bound for ED.

## B Additional Figures from Simulations

### C Metaphysics Appendix: Histories, Identity, and Free Will

We model the space of histories as a measurable space  $(\mathcal{H}, \mathcal{F})$  with prior  $P_{\text{hist}}$ . Consistency corresponds to an I-projection onto a feasible set  $\mathcal{C}$ .

**Identity.** Let  $\sigma_t$  denote a “personal” state evolving under morphisms  $U_{t \rightarrow s}$ . A retro-intervention imposes a fixed-point constraint  $U_{t \rightarrow s} U_{s \rightarrow t} \sigma_t =$

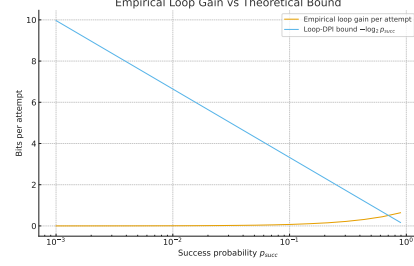


Figure 1: Empirical loop information gain per attempt versus the Loop-DPI bound as a function of postselection success probability.

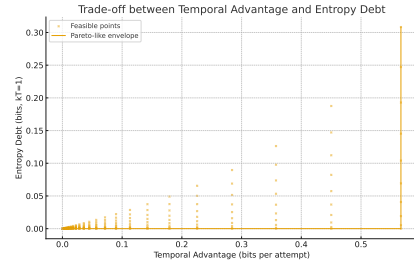


Figure 2: Pareto-like trade-off between Temporal Advantage and Entropy Debt under varying success probabilities and paradox fractions.

$\sigma_t$ . When consistency requires redistributing probability mass over narratives, the minimum change is quantified by the Entropy Ledger.

**Free Will (Operational).** Define  $\text{Free}(t) := H(A_t \mid \text{Past, Ledger Info})$ . Retro-resources can reduce Free but only by paying scarcity ( $-\log p_{\text{succ}}$ ) or heat (ED). Thus there is “no free chronology”: influence against the arrow is always priced.

## D Notation and Proof Outlines

**Loop-DPI sketch.** Normalize the non-TP retro-map  $\Pi$  by conditioning on the heralded outcome. Apply a change-of-measure argument around the cycle and bound the Radon–Nikodym derivative by  $1/p_{\text{succ}}$ , yielding  $\Delta I_{\odot} \leq -\log p_{\text{succ}}$  (bits). Finite accuracy adds a  $-\log(1 - \varepsilon)$  term.

**Entropy Ledger sketch.** Consistency is enforced by projecting  $P_{\text{hist}}$  onto  $\mathcal{C}$  in KL. The

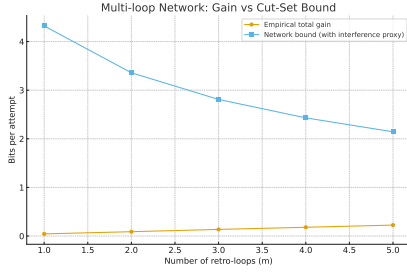


Figure 3: Multi-loop network empirical gain compared with a cut-set style bound that includes an interference proxy.

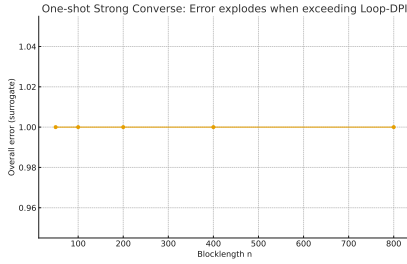


Figure 4: Surrogate demonstration of the single-shot strong converse: attempting rates above the Loop-DPI bound drives error toward one with blocklength.

work cost of modifying distributions obeys  $\langle W \rangle \geq kT D_{\text{KL}}(P \| P^*)$  by Landauer and nonequilibrium work relations.

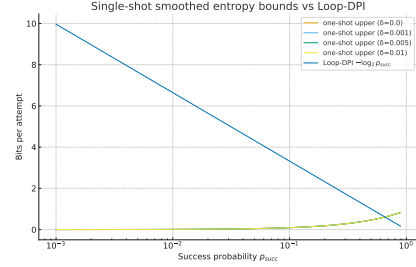


Figure 5: Single-shot (smoothed entropy) upper bounds on achievable loop bits per attempt compared to the Loop-DPI asymptote.

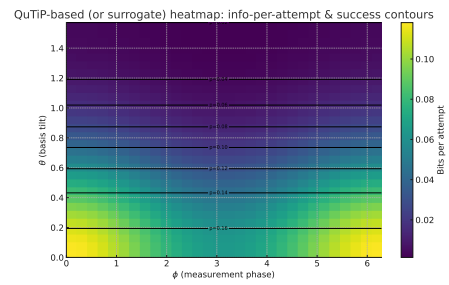


Figure 6: Heatmap of bits-per-attempt as a function of measurement basis  $(\theta, \phi)$  with success-probability contours.